

Time Series — Solutions

Practice Exam — Week 8: Spectral Estimation — Periodogram & Tapering
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Problem 1 (5 points)

Write $Y_t = X_t - \bar{X}$ throughout.

(a) Substitute the definition of $\hat{\gamma}_\tau$ into the periodogram. Only lags $|\tau| \leq n-1$ contribute, and for each such τ the inner sum runs over $t = 1, \dots, n - |\tau|$:

$$\hat{\mathcal{S}}^{(p)}(f) = \sum_{\tau=-(n-1)}^{n-1} \frac{1}{n} \sum_{t=1}^{n-|\tau|} Y_t Y_{t+|\tau|} e^{-2\pi i \tau f}.$$

The pairs $(t, t + |\tau|)$ with $1 \leq t$ and $t + |\tau| \leq n$ are exactly the pairs of indices (t, s) with $s = t + |\tau| \in T$, $t \in T$. As τ ranges over $-(n-1), \dots, n-1$ and t over the admissible range, the pair (t, s) ranges over *all* of $T \times T$ once. Setting $s = t + \tau$ for the $\tau \geq 0$ part and $s = t - |\tau|$ for the $\tau < 0$ part, we may rewrite the double sum as the free double sum over $(s, t) \in T \times T$ with $e^{-2\pi i \tau f} = e^{-2\pi i (s-t)f}$:

$$\hat{\mathcal{S}}^{(p)}(f) = \frac{1}{n} \sum_{t=1}^n \sum_{s=1}^n Y_t Y_s e^{-2\pi i (s-t)f} = \frac{1}{n} \sum_{t=1}^n \sum_{s=1}^n Y_t Y_s e^{-2\pi i s f} e^{2\pi i t f}.$$

The double sum factorises:

$$\hat{\mathcal{S}}^{(p)}(f) = \left(\sqrt{\frac{1}{n}} \sum_{t=1}^n Y_t e^{2\pi i t f} \right) \left(\sqrt{\frac{1}{n}} \sum_{s=1}^n Y_s e^{-2\pi i s f} \right) = \overline{J(f)} J(f) = |J(f)|^2,$$

since $Y_t \in \mathbb{R}$ makes $\sqrt{1/n} \sum_t Y_t e^{2\pi i t f} = \overline{J(f)}$. This is the claimed identity.

(b) *Non-negativity.* $\hat{\mathcal{S}}^{(p)}(f) = |J(f)|^2 \geq 0$ for every f , being the squared modulus of a complex number. (No SDF estimate can be negative; the representation makes this automatic.)

Evenness. Because the Y_t are real,

$$J(-f) = \sqrt{\frac{1}{n}} \sum_t Y_t e^{2\pi i t f} = \overline{J(f)},$$

hence $\hat{\mathcal{S}}^{(p)}(-f) = |J(-f)|^2 = \left| \overline{J(f)} \right|^2 = |J(f)|^2 = \hat{\mathcal{S}}^{(p)}(f)$. So $\hat{\mathcal{S}}^{(p)}$ is even.

(c) At $f = 0$ the exponential is 1, so

$$J(0) = \sqrt{\frac{1}{n}} \sum_{t=1}^n Y_t = \sqrt{\frac{1}{n}} \sum_{t=1}^n (X_t - \bar{X}) = \sqrt{\frac{1}{n}} \left(\sum_{t=1}^n X_t - n\bar{X} \right) = 0,$$

because $\sum_t X_t = n\bar{X}$ by definition of $\bar{X}$. Therefore $\hat{\mathcal{S}}^{(p)}(0) = |J(0)|^2 = 0$: removing the sample mean annihilates the zero-frequency ordinate.

Problem 2 (5 points)

Here $\mu = 0$, so $Y_t = X_t$ and $\mathbb{E}[Y_t Y_s] = \gamma_{t-s}$.

(a) Take expectations in the factorised form of the periodogram (Problem 1(a)); expectation passes through the finite double sum:

$$\mathbb{E}[\hat{\mathcal{S}}^{(p)}(f)] = \frac{1}{n} \sum_{t=1}^n \sum_{s=1}^n \mathbb{E}[Y_t Y_s] e^{-2\pi i f(t-s)} = \frac{1}{n} \sum_{t=1}^n \sum_{s=1}^n \gamma_{t-s} e^{-2\pi i f(t-s)}.$$

Collect terms by the lag $\tau = t - s$. For fixed τ with $|\tau| \leq n-1$, the number of pairs $(t, s) \in T \times T$ with $t - s = \tau$ is $n - |\tau|$ (and there are none for $|\tau| \geq n$). Hence

$$\mathbb{E}[\hat{\mathcal{S}}^{(p)}(f)] = \frac{1}{n} \sum_{\tau=-(n-1)}^{n-1} (n - |\tau|) \gamma_{\tau} e^{-2\pi i \tau f} = \sum_{\tau=-(n-1)}^{n-1} \left(1 - \frac{|\tau|}{n}\right) \gamma_{\tau} e^{-2\pi i \tau f},$$

which is the stated expression with the triangular weight $w_{\tau} = 1 - |\tau|/n$. (Equivalently, $\mathbb{E}[\hat{\gamma}_{\tau}] = (1 - |\tau|/n) \gamma_{\tau}$, then transform term by term.)

(b) With $d_t = \mathbf{1}_T(t)$ (so $d_t = 1$ for $t \in \{1, \dots, n\}$ and 0 otherwise), the autocorrelation of the window at lag τ is

$$\sum_{t \in \mathbb{Z}} d_t d_{t+|\tau|} = \#\{t : t \in T, t + |\tau| \in T\} = n - |\tau| \quad (|\tau| \leq n-1),$$

so indeed $w_{\tau} = \frac{1}{n} \sum_t d_t d_{t+|\tau|}$. Write $w = \frac{1}{n}(\tilde{d} * d)$ where $\tilde{d}_t = d_{-t}$; equivalently $w_{\tau} = \frac{1}{n}(\tilde{d} * \bar{d})_{\tau}$ with $\bar{d} = d$ (the window is real). By the discrete convolution theorem, the Fourier transform of a convolution is the product of the transforms, and the transform of $\tilde{d}$ is $\overline{D}$ where $D(f) = \sum_t d_t e^{-2\pi i t f} = \sum_{t=1}^n e^{-2\pi i t f}$. Hence the transform of w is

$$W(f) = \frac{1}{n} \overline{D(f)} D(f) = \frac{1}{n} |D(f)|^2 = \mathcal{F}_n(f),$$

the Fejér kernel. Now the expected periodogram is the transform of the product $w_{\tau} \gamma_{\tau}$; by the (dual) convolution theorem the transform of a product is the convolution of the transforms, $W * \mathcal{S}$:

$$\mathbb{E}[\hat{\mathcal{S}}^{(p)}(f)] = \sum_{\tau} w_{\tau} \gamma_{\tau} e^{-2\pi i \tau f} = \int_{-1/2}^{1/2} \mathcal{S}(f') \mathcal{F}_n(f - f') df'.$$

So the expected periodogram is the true SDF convolved with (blurred by) the Fejér kernel.

(c) Take $\mathcal{S} \equiv 1$ in the convolution identity of (b): the corresponding ACVS is $\gamma_0 = 1$, $\gamma_{\tau} = 0$ otherwise, so $\sum_{\tau} w_{\tau} \gamma_{\tau} e^{-2\pi i \tau f} = w_0 = 1$, while the right-hand side is $\int_{-1/2}^{1/2} \mathcal{F}_n(f - f') df' = \int_{-1/2}^{1/2} \mathcal{F}_n(u) du$ (periodicity, substitute $u = f - f'$). Equating,

$$\int_{-1/2}^{1/2} \mathcal{F}_n(f) df = 1.$$

(Direct check: $\int_{-1/2}^{1/2} \mathcal{F}_n = \frac{1}{n} \int |D|^2 = \frac{1}{n} \sum_{t,s} \int_{-1/2}^{1/2} e^{-2\pi i(t-s)f} df = \frac{1}{n} \sum_{t,s} \mathbf{1}_{\{t=s\}} = \frac{1}{n} \cdot n = 1$.) Since $\mathcal{F}_n \geq 0$ and integrates to 1, it is a probability density on $[-\frac{1}{2}, \frac{1}{2}]$; as $n \rightarrow \infty$ it concentrates all its mass at $f = 0$ (an approximate identity), so the convolution $\mathcal{S} * \mathcal{F}_n \rightarrow \mathcal{S}$ at every continuity point. Hence $\mathbb{E}[\hat{\mathcal{S}}^{(p)}(f)] \rightarrow \mathcal{S}(f)$: the periodogram is *asymptotically* unbiased.

Problem 3 (5 points)

Here μ is removed, $Y_t = X_t - \mu$, and we take $\mathbb{E}[Y_t Y_s] = \gamma_{t-s}$.

(a) Expand the squared modulus and take expectations:

$$\hat{\mathcal{S}}_h^{(p)}(f) = |J_h(f)|^2 = \sum_{t=1}^n \sum_{s=1}^n h_t h_s Y_t Y_s e^{-2\pi i f(t-s)},$$

$$\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \sum_{t=1}^n \sum_{s=1}^n h_t h_s \gamma_{t-s} e^{-2\pi i f(t-s)}.$$

Collect by $\tau = t - s$ (extend $h_t = 0$ off T so the t -sum is over $\mathbb{Z}$):

$$\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \sum_{\tau \in \mathbb{Z}} \left(\sum_{t \in \mathbb{Z}} h_t h_{t-\tau} \right) \gamma_{\tau} e^{-2\pi i \tau f} = \sum_{\tau \in \mathbb{Z}} w_{\tau} \gamma_{\tau} e^{-2\pi i \tau f}, \quad w_{\tau} = \sum_{t \in \mathbb{Z}} h_t h_{t+\tau},$$

where in the last step we re-indexed ($t \mapsto t + \tau$) and used that $w_{-\tau} = w_{\tau}$, so $\sum_t h_t h_{t-\tau} = \sum_t h_t h_{t+\tau}$. Now identify w as the autocorrelation of the taper: with $\tilde{h}_t = h_{-t}$ and $\bar{h}_t = h_t$,

$$(\tilde{h} * \bar{h})_{\tau} = \sum_{s \in \mathbb{Z}} \tilde{h}_{\tau-s} \bar{h}_s = \sum_s h_{s-\tau} h_s = w_{-\tau} = w_{\tau},$$

so $w = \tilde{h} * \bar{h}$. By the discrete convolution theorem its transform is the product of the transforms; the transform of $\tilde{h}$ is $\overline{H}$ and that of $\bar{h}$ is H , so

$$W(f) = \overline{H(f)} H(f) = |H(f)|^2.$$

Finally the expected tapered periodogram is the transform of the product $w_{\tau} \gamma_{\tau}$, hence the convolution of the transforms:

$$\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \int_{-1/2}^{1/2} \mathcal{S}(f') |H(f - f')|^2 df'.$$

(b) For white noise $\mathcal{S}(f') \equiv \sigma^2$ is constant, so it comes out of the integral; using periodicity and the substitution $u = f - f'$,

$$\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \sigma^2 \int_{-1/2}^{1/2} |H(f - f')|^2 df' = \sigma^2 \int_{-1/2}^{1/2} |H(u)|^2 du.$$

By Parseval for the discrete-time transform,

$$\int_{-1/2}^{1/2} |H(u)|^2 du = \int_{-1/2}^{1/2} \sum_t \sum_s h_t h_s e^{-2\pi i(t-s)u} du = \sum_t \sum_s h_t h_s \mathbf{1}_{\{t=s\}} = \sum_t h_t^2 = \|h\|_2^2,$$

using $\int_{-1/2}^{1/2} e^{-2\pi i \tau u} du = \mathbf{1}_{\{\tau=0\}}$. Therefore

$$\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \sigma^2 \|h\|_2^2.$$

The true SDF of white noise is $\mathcal{S}(f) = \sigma^2$ for all f ; the estimator is exactly unbiased, $\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \sigma^2 = \mathcal{S}(f)$ for *every* n and *every* f , precisely when the normalisation $\|h\|_2^2 = 1$ holds. (Any other normalisation would scale the estimate by the constant $\|h\|_2^2$, biasing it; this is exactly why a taper is defined with unit energy.)

(c) The multitaper estimator is the average $\hat{\mathcal{S}}^{(mt)}(f) = \frac{1}{K} \sum_{k=1}^K \hat{\mathcal{S}}_{h_k}^{(p)}(f)$. By bilinearity of the variance,

$$\text{Var}(\hat{\mathcal{S}}^{(mt)}(f)) = \frac{1}{K^2} \sum_{k=1}^K \sum_{k'=1}^K \text{Cov}(\hat{\mathcal{S}}_{h_k}^{(p)}(f), \hat{\mathcal{S}}_{h_{k'}}^{(p)}(f)).$$

Asymptotically the K tapered periodograms are uncorrelated (orthogonal tapers), so that $\text{Cov}(\hat{\mathcal{S}}_{h_k}^{(p)}, \hat{\mathcal{S}}_{h_{k'}}^{(p)}) \rightarrow \mathcal{S}(f)^2 \delta_{k,k'}$. Only the K diagonal terms survive, each $\rightarrow \mathcal{S}(f)^2$:

$$\text{Var}(\hat{\mathcal{S}}^{(mt)}(f)) \rightarrow \frac{1}{K^2} \cdot K \mathcal{S}(f)^2 = \frac{\mathcal{S}(f)^2}{K}.$$

Unlike the raw periodogram (whose variance $\rightarrow \mathcal{S}(f)^2 \neq 0$ and does *not* shrink with n), here one may let $K \rightarrow \infty$ slowly as $n \rightarrow \infty$, driving $\text{Var} \rightarrow 0$ while the bias stays controlled; averaging independent estimates is what restores consistency.

Problem 4 (5 points)

(a) Write $\theta_0 = 1$. For an MA(2), $X_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$, the ACVS is $\gamma_\tau = \sigma^2 \sum_j \theta_j \theta_{j+|\tau|}$ over the valid index range:

$$\begin{aligned}\gamma_0 &= \sigma^2(\theta_0^2 + \theta_1^2 + \theta_2^2) = \sigma^2(1 + \theta_1^2 + \theta_2^2), \\ \gamma_1 &= \sigma^2(\theta_0 \theta_1 + \theta_1 \theta_2) = \sigma^2(\theta_1 + \theta_1 \theta_2), \\ \gamma_2 &= \sigma^2 \theta_0 \theta_2 = \sigma^2 \theta_2,\end{aligned}$$

and $\gamma_\tau = 0$ for $|\tau| \geq 3$ (no overlap of the taps), with $\gamma_{-\tau} = \gamma_\tau$. Since $\{\gamma_\tau\} \in \ell^1$ the SDF is the Fourier transform of the ACVS. Writing $\gamma_\tau = \sigma^2 \sum_{j=0}^2 \sum_{k=0}^2 \theta_j \theta_k \mathbf{1}_{\{\tau=k-j\}}$,

$$\begin{aligned}\mathcal{S}(f) &= \sum_{\tau \in \mathbb{Z}} \gamma_\tau e^{-2\pi i \tau f} = \sigma^2 \sum_{j=0}^2 \sum_{k=0}^2 \theta_j \theta_k e^{-2\pi i (k-j)f} \\ &= \sigma^2 \left(\sum_{k=0}^2 \theta_k e^{-2\pi i k f} \right) \overline{\left(\sum_{j=0}^2 \theta_j e^{-2\pi i j f} \right)} = \sigma^2 \left| \sum_{k=0}^2 \theta_k e^{-2\pi i k f} \right|^2 \\ &= \sigma^2 \left| 1 + \theta_1 e^{-2\pi i f} + \theta_2 e^{-4\pi i f} \right|^2,\end{aligned}$$

the squared modulus of the MA transfer function, as claimed (and automatically ≥ 0).

(b) Expanding the squared modulus (or pairing τ with $-\tau$) gives the real cosine form

$$\mathcal{S}(f) = \gamma_0 + 2\gamma_1 \cos(2\pi f) + 2\gamma_2 \cos(4\pi f).$$

With $\theta_1 = 1$, $\theta_2 = \frac{1}{4}$, $\sigma^2 = 1$:

$$\gamma_0 = 1 + 1 + \frac{1}{16} = \frac{33}{16}, \quad \gamma_1 = 1 + \frac{1}{4} = \frac{5}{4}, \quad \gamma_2 = \frac{1}{4}.$$

Hence

$$\mathcal{S}(f) = \frac{33}{16} + \frac{5}{2} \cos(2\pi f) + \frac{1}{2} \cos(4\pi f).$$

At $f = 0$: $\cos 0 = 1$, so $\mathcal{S}(0) = \frac{33}{16} + \frac{5}{2} + \frac{1}{2} = \frac{33}{16} + \frac{48}{16} = \frac{81}{16}$. (Check: $|1 + 1 + \frac{1}{4}|^2 = (\frac{9}{4})^2 = \frac{81}{16}$.)

At $f = \frac{1}{2}$: $\cos(\pi) = -1$, $\cos(2\pi) = 1$, so $\mathcal{S}(\frac{1}{2}) = \frac{33}{16} - \frac{5}{2} + \frac{1}{2} = \frac{33}{16} - \frac{32}{16} = \frac{1}{16}$. (Check: at $f = \frac{1}{2}$, $e^{-2\pi i f} = -1$, $e^{-4\pi i f} = 1$, so $|1 - 1 + \frac{1}{4}|^2 = \frac{1}{16}$.)

Both values are positive. This is consistent with the process being a genuine MA: by part (a) the SDF is the squared modulus $\sigma^2 |\Theta(e^{-2\pi i f})|^2 \geq 0$ at *every* f , so non-negativity is automatic — exactly the property a valid SDF must have.

(c) The candidate sequence is symmetric and summable, so we may form

$$\mathcal{S}(f) = \sum_{\tau \in \mathbb{Z}} \gamma_\tau e^{-2\pi i \tau f} = \gamma_0 + \gamma_1 e^{-2\pi i f} + \gamma_{-1} e^{2\pi i f} = 1 + 2a \cos(2\pi f).$$

A sequence is a valid ACVS of some stationary process iff it is non-negative definite, equivalently (in the summable case) iff its Fourier transform satisfies $\mathcal{S}(f) \geq 0$ for all f . As f ranges over $[-\frac{1}{2}, \frac{1}{2}]$, $\cos(2\pi f)$ takes every value in $[-1, 1]$, so

$$\min_f \mathcal{S}(f) = 1 - 2|a|, \quad \max_f \mathcal{S}(f) = 1 + 2|a|.$$

The binding constraint is the minimum: $\mathcal{S}(f) \geq 0$ for all $f \iff 1 - 2|a| \geq 0 \iff |a| \leq \frac{1}{2}$.
Therefore

$$\boxed{\{\gamma_\tau\} \text{ is a valid ACVS} \iff a \in [-\frac{1}{2}, \frac{1}{2}].}$$

(This is exactly the lag-1 autocorrelation bound of an MA(1): $\rho_1 = a/\gamma_0 = a$ must lie in $[-\frac{1}{2}, \frac{1}{2}]$, attained at the boundary by the unit-coefficient MA(1) $X_t = \varepsilon_t \pm \varepsilon_{t-1}$, which has $\rho_1 = \pm\frac{1}{2}$.)